2.3.4 Engineering documents

2D drawings remain the foundational language of engineering, construction, and operations across many industries due to their ability to reduce complexity through the use of visual abstractions that convey large amounts of information through standardized symbols and conventions.

An example of this is the fragment of a *process flow diagram (PFD)* shown in figure 2.5. A PFD is commonly found in chemical or process engineering and visually maps the sequence of actions and flow of materials in a process. Through the use of standardized symbols representing a pressure vessel, valve, condenser and heater, together with conventions regarding what the lines in the diagram represent, this compact document conveys a significant amount of information to a skilled process engineer. This diagram represents the flow through a distillation column separating a mixture of pentane and hexane (the feed) into streams of pentane (distillate) and hexane (bottoms), using feedback composition control. It shows how these components are physically connected and how gases and liquids flow through the process.